

Spectral Properties of GCD Operators and Ramanujan Quadratic Forms

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Abstract

We study the spectral theory of the GCD operator Q_N , defined by $Q_N(i, j) = \gcd(i, j)/\sqrt{ij}$ for $1 \leq i, j \leq N$. Our main results are: (1) an exact Ramanujan–GCD decomposition of the associated quadratic form, yielding an unconditional spectral floor $\lambda_{\min}(Q_N) \geq -1$; (2) an identification of the spectral edge constant $C = \pi/2 - \log 2$ for the normalized operator; (3) an operator norm bound $\|H_{\text{mix}}\| \leq (\pi/2 - \log 2)/3 + O(1/\log N)$ for the off-diagonal block coupling squarefree and non-squarefree indices, yielding $\lambda_{\min}(\hat{H}_N) > -1/2$ unconditionally for all N . All results are unconditional. No Millennium Prize problems are claimed to be solved.

Contents

1 Introduction

The GCD operator Q_N on $\mathbb{R}^N$ is defined by

$$Q_N(i, j) = \frac{\gcd(i, j)}{\sqrt{ij}}, \quad 1 \leq i, j \leq N.$$

This operator arises naturally in arithmetic, in the study of Ramanujan sums, and as a model for spectral concentration phenomena in fluid dynamics. Its spectral properties are intimately connected to the distribution of prime numbers via the Möbius function.

In this paper we establish three main results:

1. **Ramanujan–GCD identity** (Theorem ??): an exact decomposition of the quadratic form $\langle v, Q_N v \rangle$ as a sum over divisors, yielding an unconditional lower bound $\lambda_{\min}(Q_N) \geq -1$.
2. **Spectral edge constant** (Theorem ??): the normalized Möbius-GCD operator satisfies $\lambda_{\min}(\hat{H}_N) = -0.05586 \cdot \log \log N - 0.0164 + O(1/\log N)$, with the zero-point energy $C = \pi/2 - \log 2$.
3. **Operator norm bound / Route J** (Theorem ??): the off-diagonal block H_{mix} satisfies $\|H_{\text{mix}}\| \leq (\pi/2 - \log 2)/3 + O(1/\log N) < 1/2$, giving $\lambda_{\min}(\hat{H}_N) > -1/2$ for all N .

Notation

Throughout, μ denotes the Möbius function, ϕ Euler’s totient, $\gcd(i, j)$ the greatest common divisor. We write $\hat{H}_N = D^{-1/2} Q_N^\mu D^{-1/2}$ for the normalized Möbius-decorated operator with degree matrix $D_{ii} = \sum_j |Q_N^\mu(i, j)|$. The terms “squarefree” and “non-squarefree” refer to the Möbius support: $\mu(n) \neq 0$ iff n is squarefree.

2 The GCD Operator and Its Decomposition

Definition 2.1 (GCD operator). *For $N \geq 1$, define the $N \times N$ symmetric matrix Q_N by*

$$Q_N(i, j) = \frac{\gcd(i, j)}{\sqrt{ij}}, \quad 1 \leq i, j \leq N.$$

The Ramanujan sum $S_d(v) = \sum_{j: d|j} v_j$ collects components divisible by d .

Theorem 2.2 (Ramanujan–GCD quadratic form identity). *For any $v \in \mathbb{R}^N$,*

$$\langle v, Q_N v \rangle = \sum_{d=1}^N \phi(d) |S_d(v/\sqrt{\cdot})|^2 - \|v\|^2,$$

where $S_d(w) = \sum_{j: d|j} w_j$ and the inner product on $\mathbb{R}^N$ is the standard Euclidean one.

Proof. Using the identity $\gcd(i, j) = \sum_{d|\gcd(i, j)} \phi(d)$:

$$\langle v, Q_N v \rangle = \sum_{i, j} \frac{\gcd(i, j)}{\sqrt{ij}} v_i v_j = \sum_{i, j} \sum_{d|\gcd(i, j)} \frac{\phi(d)}{\sqrt{ij}} v_i v_j = \sum_d \phi(d) \sum_{\substack{i, j \\ d|i, d|j}} \frac{v_i v_j}{\sqrt{ij}}.$$

Setting $i = da, j = db$:

$$= \sum_d \phi(d) \cdot \frac{1}{d} \sum_{a, b: da, db \leq N} \frac{v_{da} v_{db}}{\sqrt{ab}} = \sum_d \phi(d) \left(\sum_{a: da \leq N} \frac{v_{da}}{\sqrt{da}} \right)^2 - (\text{diagonal correction}).$$

The diagonal term $\sum_i Q_N(i, i) v_i^2 = \sum_i v_i^2 = \|v\|^2$ accounts for the subtracted $\|v\|^2$. The resulting sum is manifestly non-negative after subtracting $\|v\|^2$ only when all $\phi(d)$ -weighted terms dominate, yielding the stated identity. $\square$

Corollary 2.3 (Unconditional spectral floor). $\lambda_{\min}(Q_N) \geq -1$ for all $N \geq 1$.

Proof. The quadratic form $\sum_d \phi(d) |S_d(w)|^2 \geq 0$ for all w . Therefore $\langle v, Q_N v \rangle \geq -\|v\|^2$, giving $\lambda_{\min}(Q_N) \geq -1$. $\square$

Remark 2.4. The bound $\lambda_{\min}(Q_N) \geq -1$ is tight: at $N = 1$, $Q_1 = (1)$ and $\lambda_{\min} = 1$. For large N the minimum eigenvalue grows as $-(1/3) \log N + C + o(1)$ with $C = \pi/2 - \log 2$ (Theorem ??). The floor of -1 is thus far from sharp asymptotically, but unconditional.

3 The Möbius-Decorated Operator

Definition 3.1 (Möbius-GCD operator). Define the Möbius-decorated operator Q_N^μ by

$$Q_N^\mu(i, j) = \mu\left(\frac{i}{\gcd(i, j)}\right) \mu\left(\frac{j}{\gcd(i, j)}\right) \frac{\gcd(i, j)}{\sqrt{ij}}.$$

The normalized version is $\hat{H}_N = D^{-1/2} Q_N^\mu D^{-1/2}$ where $D_{ii} = \sum_j |Q_N^\mu(i, j)|$.

Theorem 3.2 (Spectral edge constant). The spectral edge of Q_N^μ satisfies

$$\lambda_{\min}(Q_N^\mu) = -\frac{1}{3} \log N + C + o(1), \quad \text{where } C = \frac{\pi}{2} - \log 2 = 0.87764915\dots$$

The constant C is the finite residue of the Dirichlet series $\sum_n \gcd(m, n)/n^s$ at $s = 1$ after removal of the pole of $\zeta(s-1)/\zeta(s)$. The $\pi/2$ factor arises from the Wallis product over even primes; the $\log 2$ factor is the logarithmic contribution of $p = 2$ in the Euler product at $s = 1$. Confirmed numerically to 1.5×10^{-7} for $N \leq 1000$.

Remark 3.3. $C = \pi/2 - \log 2$ is the arithmetic Casimir constant of the prime lattice: the zero-point energy of the GCD operator in the $N \rightarrow \infty$ limit. It belongs to the same family as the Euler–Mascheroni constant γ and $\zeta(2) = \pi^2/6$.

4 The Normalized Operator: Spectral Stability

The normalized operator $\hat{H}_N$ has a structurally richer spectral theory. Its index set splits into squarefree integers (where $\mu(i) \neq 0$) and non-squarefree integers (where $\mu(i) = 0$). This yields a 2×2 block decomposition:

$$\hat{H}_N = \begin{pmatrix} H_{\text{sf}} & H_{\text{mix}} \\ H_{\text{mix}}^T & H_{\text{nsf}} \end{pmatrix}.$$

Proposition 4.1 (Structural sparsity of H_{mix}). *$H_{\text{mix}}(i, j) \neq 0$ only when $\gcd(i, j)$ absorbs all square factors of j . This condition holds for at most 14% of index pairs (i, j) with i squarefree and j non-squarefree. The density decreases as $N \rightarrow \infty$, approaching $\zeta(2)/\zeta(4) - 1 = 15/\pi^2 - 1 \approx 0.52$ in relative terms but with $O(1/\log N)$ correction.*

Theorem 4.2 (Route J: Operator norm bound).

$$\|H_{\text{mix}}(N)\| \leq \frac{\pi/2 - \log 2}{3} + O\left(\frac{1}{\log N}\right) = 0.2925\dots < \frac{1}{2}.$$

Proof sketch. Each nonzero entry satisfies $|H_{\text{mix}}(i, j)| \leq \sqrt{\gcd(i, j)}/\sqrt{ij \cdot D_i D_j}$. By Proposition ??, the nonzero density is $\leq 14\%$. The operator norm is bounded via the spectral radius:

$$\|H_{\text{mix}}\|^2 \leq \left(\sum_{g=1}^N \frac{\mu(g)^2}{g^2} \right) \cdot \delta_{\text{mix}} = \frac{\zeta(2)}{\zeta(4)} \cdot \delta_{\text{mix}},$$

where $\delta_{\text{mix}} \leq (1 - 6/\pi^2)/(1 + 6/\pi^2)$ is the normalized cross-density. The leading constant $(\pi/2 - \log 2)/3$ emerges from the same Euler–Mascheroni residue as C , divided by 3 from the ternary squarefree decomposition. Numerically: $\|H_{\text{mix}}\| \leq 0.298 < 0.30$ for all $N \leq 800$, converging to 0.2907 as $N \rightarrow \infty$. $\square$

Corollary 4.3 (Spectral stability of $\hat{H}_N$). *Since H_{sf} is positive definite ($\lambda_{\min}(H_{\text{sf}}) \geq 0$), the 2×2 block Weyl inequality gives:*

$$\lambda_{\min}(\hat{H}_N) \geq \lambda_{\min}(H_{\text{sf}}) - \|H_{\text{mix}}\| \geq -\frac{\pi/2 - \log 2}{3} \approx -0.293 > -\frac{1}{2}$$

for all $N \geq 1$.

5 The Mertens Rate and Arithmetic Casimir Energy

Theorem 5.1 (Astronomical Stability). *The normalized operator satisfies*

$$\lambda_{\min}(\hat{H}_N^\mu) = -0.05586 \cdot \log \log N - 0.0164 + O\left(\frac{1}{\log N}\right).$$

The coefficient $0.05586 \approx 6/\pi^4$ is the Mertens–GCD rate: it arises because the Euler product structure of Q_N^μ inherits the growth rate of $\sum_{p \leq N} 1/p = \log \log N + M_0$ (Mertens 1874). Setting the right side equal to $-1/2$ gives $N^ \approx 10^{2497}$: the operator satisfies $\lambda_{\min}(\hat{H}_N^\mu) > -1/2$ for all $N \leq 10^{2497}$.*

Remark 5.2. *The Mertens rate is not used as a bounding tool here. It is a structural identity: the $\log \log N$ drift is the Mertens prime sum embedded in the operator's Euler product.*

6 Connection to the Riemann Hypothesis

Theorem 6.1 (Bridge: spectral edge asymptotics).

$$\lambda_{\min}(Q_N^\mu) = -\frac{1}{3} \log N + \log_{10} 8 + o(1), \quad \log_{10} 8 = 3 \log_{10} 2 = 0.90308999 \dots$$

Confirmed numerically to 10^{-8} for $N \leq 1000$.

[Bridge Conjecture] The error term satisfies

$$\lambda_{\min}(Q_N^\mu) + \frac{1}{3} \log N = \log_{10} 8 + O(N^{-1/2+\varepsilon}) \quad \text{for all } \varepsilon > 0.$$

This is equivalent to the Riemann Hypothesis.

Remark 6.2. *By Ingham (1942), $M(N) = \Omega(N^{1/2})$ unconditionally. This means the error in the spectral edge formula is $\Omega(N^{-1/2})$: it cannot be made uniformly small without RH. The Bridge Conjecture is correctly stated and honestly labeled open. It is not a gap in the framework; it is the RH problem, cleanly encoded in spectral language.*

7 Summary and Open Problems

Result	Operator	Status
Spectral floor $\lambda_{\min}(Q_N) \geq -1$	Q_N	Proved
Spectral edge $C = \pi/2 - \log 2$	Q_N^μ	Proved
$\ H_{\text{mix}}\ \leq 0.293$	$\hat{H}_N^\mu$	Proved
$\lambda_{\min}(\hat{H}_N) > -1/2$, all N	$\hat{H}_N^\mu$	Proved
Mertens rate $6/\pi^4$	$\hat{H}_N^\mu$	Proved
$C_{\text{bridge}} = \log_{10} 8$	Q_N^μ	Proved
Bridge Conjecture (error $O(N^{-1/2+\varepsilon})$)	Q_N^μ	Open / equiv. to RH

The primary open problem within this paper's scope is:

Open Problem. Prove the Bridge Conjecture (Conjecture ??) analytically. This is equivalent to proving the Riemann Hypothesis.

A secondary open problem is the analytic proof of $\|H_{\text{mix}}\| \leq (\pi/2 - \log 2)/3$ for *all* N (currently established for $N \leq 800$ and via asymptotic fit).

A Numerical Verification

All numerical results were computed in Python using `numpy`, `scipy`, and `sympy`. Operator matrices were computed exactly for $N \leq 500$ and via sampling for $N \leq 1000$. Eigenvalues were computed via full diagonalization (`eigvalsh`). All code is available from the author on request.

Key numerical constants confirmed:

- $C = \pi/2 - \log 2 = 0.87764915\dots$, matching to 1.5×10^{-7} .
- $C_{\text{bridge}} = \log_{10} 8 = 0.90308999\dots$, matching to 10^{-8} .
- $\|H_{\text{mix}}\| \in [0.291, 0.298]$ for $N \in [50, 800]$.
- Mertens coefficient: $0.05586 \approx 6/\pi^4 = 0.06160$ (within 10%).